

Robust Direction-Preserving Control Allocation

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This paper investigates robust linear control allocation for aircraft with parametric model uncertainty. We augment a standard linear programming-based allocator with a robustified direction preserving constraint, which ensures that the realized generalized moment is approximately collinear with the command over the specified uncertainty set. This results in a second-order cone program which can be solved reliably in real time. Using an example with the F-18 high-alpha research vehicle, the proposed allocator is compared to similar methods and demonstrated to best preserve realized moment directions under uncertainty while being able to realize nearly all physically-attainable moments.

Nomenclature

x	vehicle states
u	control inputs
$\underline{u}$	control effector lower limits
$\overline{u}$	control effector upper limits
u_0	preferred controls
τ	generalized moments
τ_c	generalized moment commands
B	control effectiveness matrix
B_0	nominal control effectiveness matrix
B_Δ^k	control effectiveness uncertainty matrices
Δ_B	control effectiveness uncertainty set
c_1, c_2	control allocation weighting terms
p, q, r	body-axis angular rates, deg/s
u_{DT}	differential tail, deg
u_A	aileron, deg
u_R	rudder, deg
u_{RTV}	roll thrust vectoring, deg
u_{YTV}	yaw thrust vectoring, deg
$[\cdot]_i$	i th row of matrix or vector

I. Introduction

TRADITIONAL aircraft control design has typically dealt with three rotational degrees of freedom and three corresponding control inputs. However, modern aircraft generally have far more control effectors than degrees of freedom. These control effectors include independently operating aerodynamic control surfaces and thrust-vectoring moment generators, among others, and can affect translational as well as rotational degrees of freedom. As control design with so many degrees of freedom and cross-coupled dynamics is challenging, it is desirable to develop control allocation systems to map generalized moment commands (e.g., desired angular accelerations or translational forces) to individual control effector commands.

Generalized inverse (GI) control allocators are the most common class of control allocation approaches. GI allocation uses a closed-form linear transformation derived from a local linearization of the control effectiveness. The simplest GI is a Moore-Penrose pseudo-inverse (PI) allocator. Although practical for real-time implementation, in general GI

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allocators cannot access all physically attainable generalized moments when there are constraints on effector positions. This inability to realize all possible generalized moments can limit flight vehicle performance [1].

Control allocation is also commonly formulated as a linear programming (LP) problem and is solved online using convex optimization solvers [2–6]. The main advantage of LP methods is that nearly all physically attainable generalized moment commands can be realized. Different formulations of the LP problem will prioritize moment collinearity or reducing magnitude error, resulting in differences in performance when commanded generalized moments are unattainable. Typically, control laws are not constrained to command only attainable generalized moments, and unattainable generalized moment commands can be expected during flight [1]. Consequently, different LP allocation formulations can result in differences in stability and piloted handling qualities [7].

Several methods have been derived to solve the control allocation problem in the presence of parametric model uncertainty. For example, Ref. [8] derives the minimum-variance (MV) allocator, which is a special case of GI designed to minimize the variance of error in generalized moment. Reference [9] derives a robust version of the minimum magnitude error LP allocator from [5] for ellipsoidal uncertainty sets. The robustified LP allocator is formulated as a second-order cone programming (SOCP) problem, which is a class of convex optimization problems with low complexity and which can be solved in real-time [10–15].

The contribution of this paper is to introduce the *robust direction-preserving cone programming* (RDPCP) allocator. Like the robust minimum error allocator, RDPCP is formulated as a SOCP. The proposed allocator is designed to achieve a large portion of attainable generalized moments, while maintaining collinearity with commanded moments for a specified level of uncertainty. This robust direction-preserving property can mitigate unintended cross-axis coupling between moment axes, thus potentially preserving stability margins and flying qualities metrics.

The remainder of this paper is organized as follows. Section I.A reviews the control allocation problem and section I.B describes the uncertainty model considered in this work. Section II reviews the derivation of existing LP allocators and the robust minimum error allocator. Next, section III derives the proposed RDPCP allocator. Section IV studies the efficacy of the RDPCP allocator compared to other common allocation approaches. Lastly, section V provides concluding remarks and recommendations for future research.

A. The Control Allocation Problem

In general, flight vehicle equations of motion can be written as

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\tau}) \quad (1a)$$

$$\boldsymbol{\tau} = \mathbf{h}(\mathbf{x}, \mathbf{u}) \quad (1b)$$

where, $\mathbf{x} \in \mathbb{R}^{n_x}$ is the state, $\mathbf{u} \in \mathbb{R}^{n_u}$ is the control input, $\boldsymbol{\tau} \in \mathbb{R}^{n_\tau}$ represents generalized moments resulting from vehicle states and effectors, $\mathbf{f} : \mathbb{R}^{n_x} \times \mathbb{R}^{n_\tau} \rightarrow \mathbb{R}^{n_x}$ is the vehicle dynamics, and $\mathbf{h} : \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \rightarrow \mathbb{R}^{n_\tau}$ is the generalized moment model. The generalized moments $\boldsymbol{\tau}$ typically include moments, angular accelerations, and forces. Throughout this paper we will refer to generalized moments simply as moments when the meaning is unambiguous. The control effectors are required to satisfy box constraints,

$$\underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}}, \quad (2)$$

for lower and upper limits $\underline{\mathbf{u}}, \bar{\mathbf{u}} \in \mathbb{R}^{n_u}$, respectively. Note that control effector rate limits can also be imposed by tightening the box constraints around the current effector position at each instant in time [1].

We consider the typical case where the vehicle control effectors appear linearly in the equations of motion (or the allocation design model is based on linearization), that is,

$$\boldsymbol{\tau} = \mathbf{B}\mathbf{u}, \quad \mathbf{B} \in \mathbb{R}^{n_\tau \times n_u}. \quad (3)$$

For an allocation solution to exist it is assumed that $\mathbf{B}$ has full row rank, or in other words, the control inputs have independent authority over each moment axis. A control system commands a moment, $\boldsymbol{\tau}_c \in \mathbb{R}^{n_\tau}$, to be imparted by the vehicle control effectors. The control allocation problem is to find a mapping between the moment commands and the control effectors, i.e., find an admissible $\mathbf{u}$ such that $\boldsymbol{\tau} = \boldsymbol{\tau}_c$.

We say that a moment command $\boldsymbol{\tau}_c$ is *attainable* if there exists a control input $\mathbf{u}$ with $\underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}}$ such that $\boldsymbol{\tau} = \boldsymbol{\tau}_c$. When the moment command is unattainable (i.e., no feasible $\mathbf{u}$ satisfies $\mathbf{B}\mathbf{u} = \boldsymbol{\tau}_c$) or the allocator fails to find a feasible solution, this results in an allocation error. In this paper we decompose allocation error into *magnitude error* and *direction error*. We use the following relative error metrics:

$$\text{magnitude error} := \frac{\|\mathbf{B}\mathbf{u} - \boldsymbol{\tau}_c\|_1}{\|\boldsymbol{\tau}_c\|_1}, \quad \text{direction error} := \cos^{-1} \left(\frac{\boldsymbol{\tau}_c^T \mathbf{B}\mathbf{u}}{\|\boldsymbol{\tau}_c\|_2 \|\mathbf{B}\mathbf{u}\|_2} \right). \quad (4)$$

Note that when a denominator in the relative metrics is small, i.e., less than 10^{-6} , we use a small positive constant to avoid numerical problems.

Typically $n_u > n_\tau$, hence there can be infinitely many feasible solutions to the control allocation problem. Even when a moment is unattainable, different control allocations can yield significant differences in performance. Because of this, many control allocation methods have been developed. The performance of an allocator for a general problem* can be judged by several properties:

- Magnitude and direction error should be negligible for a large portion of attainable moments.
- Allocation error should grow smoothly and predictably beyond the limit of attainable moments. An allocation scheme could further choose to prioritize minimizing the magnitude error or direction error of unattainable moments. Such approaches are referred to as minimum error and direction-preserving allocators, respectively. The choice between these two approaches could depend on stability, performance, or piloted handling qualities evaluations, among other criteria.
- Allocation solutions should also minimize deviation of control effectors from a preferred position.
- These properties should be robust to uncertainty in $\mathbf{B}$ within a specified range, and degrade gracefully for model errors outside of this range.

B. Uncertainty Model

Importantly, Eq. (3) depends on a model of $\mathbf{B}$. Such linearized models are based on physics models and/or observed data, and so we should expect them to have some inaccuracy. This motivates the use of a robust allocator, whose performance will be able to tolerate perturbations from the nominal model. In this paper, we will consider ellipsoidal uncertainty sets in $\mathbf{B}$, as these can capture a wide variety of uncertainty models and lead to tractable robust optimization properties.

Let $\mathbf{B}_0$ be the nominal value of $\mathbf{B}$, and let $\mathbf{B}_\Delta^k \in \mathbb{R}^{n_\tau \times n_u}$, $k = 1, \dots, n_\Delta$ be matrices describing maximal perturbations to $\mathbf{B}_0$ within the uncertainty model. Then we define the uncertainty set by the ellipsoid

$$\Delta_B := \left\{ \mathbf{B}_0 + \sum_{k=1}^{n_\Delta} \delta_k \mathbf{B}_\Delta^k \left| \sum_{k=1}^{n_\Delta} \delta_k^2 \leq 1 \right. \right\}. \quad (5)$$

As a simple example, we can consider independent variations in each entry of $\mathbf{B}$. Let $\sigma_{i,j} \geq 0$ be the maximum uncertainty (e.g., two standard deviations) of the i, j th entry of $\mathbf{B}$, and let $\mathbf{e}_i^n$ denote the i th unit vector in $\mathbb{R}^n$. Then we can define perturbation matrices $\mathbf{B}_\Delta^{i,j} = \sigma_{i,j} \mathbf{e}_i^{n_\tau} (\mathbf{e}_j^{n_u})^T$ for $i = 1, \dots, n_\tau$ and $j = 1, \dots, n_u$. That is, $\mathbf{B}_\Delta^{i,j}$ is a matrix of all zeros, except for the i, j th entry which equals $\sigma_{i,j}$. The ellipsoid uncertainty set can be expressed as

$$\Delta_B := \left\{ \mathbf{B}_0 + \sum_{i=1}^{n_\tau} \sum_{j=1}^{n_u} \delta_{i,j} \sigma_{i,j} \mathbf{e}_i^{n_\tau} (\mathbf{e}_j^{n_u})^T \left| \sum_{i=1}^{n_\tau} \sum_{j=1}^{n_u} \delta_{i,j}^2 \leq 1 \right. \right\}. \quad (6)$$

II. Review of Control Allocation by Linear Programming

In this section, we review two standard LP-based control allocation methods, as well as an existing robust SOCP-based method which minimizes magnitude error. These convex programming allocation formulations can be parsed into standard-form LP and SOCP problems as described in Refs. [5, 9]. This allows one to solve them in real time using mature simplex or interior-point-based algorithms available in commercial software [10–15].

A. Direction-Preserving Formulation (DPLP)

A common formulation of the control allocation problem is to satisfy the constraint $\boldsymbol{\tau}_c = \mathbf{B}\mathbf{u}$ exactly when possible, and, when this is not possible, preserve the *direction* of the command. We will refer to this approach as direction-preserving LP (DPLP), following Ref. [1].

The DPLP allocator is formulated as follows. We introduce the auxiliary variable $0 \leq \lambda \leq 1$, which we will seek to

*In addition to these criteria, some vehicle-specific allocation objectives can include minimizing control energy, drag, or internal loads.

maximize, and the constraint $\mathbf{B}\mathbf{u} = \lambda\boldsymbol{\tau}_c$. This gives us the following LP:

$$\min_{\mathbf{u}, \lambda} \quad -\lambda \quad (7a)$$

$$\text{s.t.} \quad \underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}} \quad (7b)$$

$$\mathbf{B}\mathbf{u} = \lambda\boldsymbol{\tau}_c \quad (7c)$$

$$0 \leq \lambda \leq 1 \quad (7d)$$

Note that in the trivial case $\boldsymbol{\tau}_c = \mathbf{0}$, this problem is not well-defined, so one sets $\mathbf{u} = \mathbf{0}$ (see Ref. [5]). Observe that when $\boldsymbol{\tau}_c$ is attainable, the solution to Eqs. (7a)–(7d) will satisfy $\lambda = 1$ and $\boldsymbol{\tau}_c = \mathbf{B}\mathbf{u}$. Otherwise, $\lambda < 1$ so $\mathbf{B}\mathbf{u}$ will be collinear with $\boldsymbol{\tau}_c$, but scaled such that $\mathbf{u}$ does not exceed the constraints.

B. Minimum ℓ_1 Error Formulation (ℓ_1 LP)

A second common formulation is to minimize a weighted sum of the ℓ_1 norm of the allocation error and the deviation from a preferred control vector [5]. We refer to this approach as ℓ_1 LP.

The LP problem is set up as the following mixed-objective optimization problem [5]:

$$\min_{\mathbf{u}} \quad \|\mathbf{B}\mathbf{u} - \boldsymbol{\tau}_c\|_1 + c_1 \|\mathbf{u} - \mathbf{u}_0\|_1 \quad (8a)$$

$$\text{s.t.} \quad \underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}} \quad (8b)$$

Here $\mathbf{u}_0$ represents a preferred control input with $\underline{\mathbf{u}} \leq \mathbf{u}_0 \leq \bar{\mathbf{u}}$, and $c_1 \geq 0$ is a small weighting parameter which allows the control designer to prioritize solutions which are close to preferred control inputs or minimize magnitude error.

C. Robust Minimum ℓ_1 Error Formulation ($\text{R}\ell_1$ CP)

More recently, Ref. [9] proposed a robust form of the certain ℓ_1 LP problem for the case of ellipsoidal uncertainty in $\mathbf{B}$. The idea is to try to solve Eqs. (8a) and (8b) for all possible variations of the control effectiveness in the parametric ellipsoid uncertainty set, Δ_B , described in Eq. (5). To make this problem tractable, the method presented in Ref. [16] is used to derive a *robust counterpart* which depends only on the worst-case perturbations from the nominal $\mathbf{B}_0$. We will refer to this as method as robust ℓ_1 cone programming ($\text{R}\ell_1$ CP).

To derive the $\text{R}\ell_1$ CP problem, Ref. [9] introduces the slack variable $\mathbf{s} \in \mathbb{R}^{n_\tau}$ to rewrite Eqs. (8a) and (8b) as

$$\min_{\mathbf{u}, \mathbf{s}} \quad \|\mathbf{s}\|_1 + c_1 \|\mathbf{u} - \mathbf{u}_0\|_1 \quad (9a)$$

$$\text{s.t.} \quad \underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}} \quad (9b)$$

$$\mathbf{0} \leq \mathbf{s} \quad (9c)$$

$$-\mathbf{s} \leq \mathbf{B}\mathbf{u} - \boldsymbol{\tau}_c \leq \mathbf{s} \quad (9d)$$

The uncertain LP replaces the constraint Eq. (9d) with

$$-\mathbf{s} \leq \mathbf{B}\mathbf{u} - \boldsymbol{\tau}_c \leq \mathbf{s}, \quad \forall \mathbf{B} \in \Delta_B \quad (10)$$

Instead of trying to solve this problem for all possible perturbations (an intractable problem), we consider the worst-case perturbations in the convex set Δ_B . This is called the robust counterpart to the uncertain LP. Because the uncertainty set Δ_B is ellipsoidal, one can show that the robust counterpart is the following SOCP [9, 16]:

$$\min_{\mathbf{u}, \mathbf{s}} \quad \|\mathbf{s}\|_1 + c_1 \|\mathbf{u} - \mathbf{u}_0\|_1 \quad (11a)$$

$$\text{s.t.} \quad \underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}} \quad (11b)$$

$$\mathbf{0} \leq \mathbf{s} \quad (11c)$$

$$\sqrt{\sum_{k=1}^{n_\Delta} \left([\mathbf{B}_\Delta^k]_i \mathbf{u} \right)^2} \leq s_i - [\mathbf{B}_0]_i \mathbf{u} + [\boldsymbol{\tau}_c]_i, \quad i = 1, \dots, n_\tau \quad (11d)$$

$$\sqrt{\sum_{k=1}^{n_\Delta} \left([\mathbf{B}_\Delta^k]_i \mathbf{u} \right)^2} \leq s_i + [\mathbf{B}_0]_i \mathbf{u} - [\boldsymbol{\tau}_c]_i, \quad i = 1, \dots, n_\tau \quad (11e)$$

III. Robust Direction-Preserving Control Allocation

In this section we describe the proposed RDPCP allocator. To derive RDPCP, we first augment the standard DPLP formulation (Eqs. (7a)–(7d)) with a bias term to encourage solutions near a preferred control. This has the dual benefit of regularizing the problem and reducing control effort. Then we derive a robust counterpart to this allocation problem, which helps preserve moment direction over a specified model uncertainty space. The addition of a preferred control and robustification process results in a SOCP-based allocation problem which can be solved in real time by a conic optimization solver with guaranteed convergence properties [10–15].

A. Robust Direction-Preserving Formulation with Preferred Control (RDPCP)

First we describe the certain allocation problem, in which we augment the standard DPLP Eqs. (7a)–(7d) with a preferred control bias term like in Eq. (8a). This results in another LP:

$$\min_{\mathbf{u}, \lambda} \quad c_1 \|\mathbf{u} - \mathbf{u}_0\|_1 - c_2 \lambda \quad (12a)$$

$$\text{s.t.} \quad \underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}} \quad (12b)$$

$$0 \leq \lambda \leq 1 \quad (12c)$$

$$\mathbf{B}\mathbf{u} = \lambda \boldsymbol{\tau}_c \quad (12d)$$

Here $c_1, c_2 > 0$ in Eq. (12a) are weighting parameters which affect the tradeoff between minimizing deviation from the preferred control and maximizing the magnitude of the realized moment. Although the two parameters are technically redundant in this problem, including both allows the same value of c_1 to be used for different allocators. Furthermore, both c_1 and c_2 will be used in the robustified problem.

We now incorporate uncertainty in the control effectiveness matrix, $\mathbf{B}$. We would like to minimize Eq. (12a) subject to Eqs. (12b)–(12d), for all $\mathbf{B}$ in the ellipsoid uncertainty set Δ_B in Eq. (5). Of course, the same control vector cannot satisfy the equality constraint Eq. (12d) for all possible $\mathbf{B}$, so we introduce the slack variable $\mathbf{s} \in \mathbb{R}^{n_\tau}$ and rewrite the equality constraint as a set of inequalities:

$$\min_{\mathbf{u}, \lambda, \mathbf{s}} \quad \|\mathbf{s}\|_1 + c_1 \|\mathbf{u} - \mathbf{u}_0\|_1 - c_2 \lambda \quad (13a)$$

$$\text{s.t.} \quad \underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}} \quad (13b)$$

$$0 \leq \lambda \leq 1 \quad (13c)$$

$$\mathbf{0} \leq \mathbf{s} \quad (13d)$$

$$-\mathbf{s} \leq \mathbf{B}\mathbf{u} - \lambda \boldsymbol{\tau}_c \leq \mathbf{s}, \quad \forall \mathbf{B} \in \Delta_B \quad (13e)$$

To make the uncertain LP Eqs. (13a)–(13e) tractable, we apply [16, Theorem 3.1] to derive a robust counterpart which considers the worst-case uncertainty. For the ellipsoidal uncertainty set Δ_B , this leads to the following SOCP:

$$\min_{\mathbf{u}, \lambda, \mathbf{s}} \quad \|\mathbf{s}\|_1 + c_1 \|\mathbf{u} - \mathbf{u}_0\|_1 - c_2 \lambda \quad (14a)$$

$$\text{s.t.} \quad \underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}} \quad (14b)$$

$$0 \leq \lambda \leq 1 \quad (14c)$$

$$\mathbf{0} \leq \mathbf{s} \quad (14d)$$

$$\sqrt{\sum_{k=1}^{n_\Delta} \left([\mathbf{B}_\Delta^k]_i \mathbf{u} \right)^2} \leq s_i - [\mathbf{B}_0]_i \mathbf{u} + \lambda [\boldsymbol{\tau}_c]_i, \quad i = 1, \dots, n_\tau \quad (14e)$$

$$\sqrt{\sum_{k=1}^{n_\Delta} \left([\mathbf{B}_\Delta^k]_i \mathbf{u} \right)^2} \leq s_i + [\mathbf{B}_0]_i \mathbf{u} - \lambda [\boldsymbol{\tau}_c]_i, \quad i = 1, \dots, n_\tau \quad (14f)$$

Comparing the proposed RDPCP allocator Eqs. (14a)–(14f) with the $\mathcal{R}\ell_1$ CP allocator Eqs. (11a)–(11e), notice that $\mathcal{R}\ell_1$ CP can be recovered by fixing $\lambda \equiv 1$. Indeed, one could also obtain Eqs. (14a)–(14f) from Eqs. (11a)–(11e) by an ad hoc introduction of λ , but we prefer deriving it by robustifying the modified DPLP allocator (Eqs. (12a)–(12d)). The addition of the free variable λ allows RDPCP to prioritize robust direction preservation for unattainable moment commands.

As with ℓ_1 LP and $R\ell_1$ CP, c_1 is a small weighting parameter which encourages solutions near $\mathbf{u}_0$. In numerical experiments, we will use the same value of c_1 for ℓ_1 LP, $R\ell_1$ CP, and RDPCP.

In practice, setting $c_2 \gg \|\tau_c\|$ yields similar behavior as $R\ell_1$ CP. On the other hand, if we choose $c_2 \ll \|\tau_c\|$, then the solution tends to $\mathbf{u} \rightarrow \mathbf{u}_0$ and $\tau \rightarrow \mathbf{0}$, even for non-zero moment commands (see section III.B). In general, making c_2 large prioritizes preserving the *magnitude* of the realized moment, whereas making c_2 small prioritizes reducing the slack, thus robustly preserving the *direction* of the realized moment. We find that an appropriate choice of $c_2 = O(\|\tau_c\|)$ prioritizes robust direction preservation while still realizing a large percentage of attainable moments.

B. Bounds on c_1 and c_2 Parameters

Here we present a brief analysis to suggest guidelines for selecting parameters, c_1 and c_2 , in Eqs. (14a)–(14f). In early experiments we found that choosing c_2 too small led to decay of solutions for large commands and uncertainty. That is, when c_2 is small and uncertainty is large, commanding a large moment results in zero control output. That is, we observed $\lambda \rightarrow 0$, $\mathbf{u} \rightarrow \mathbf{u}_0$, and $\tau \rightarrow \mathbf{0}$, when $c_2 \ll \|\tau_c\|$. This motivates the search for sufficient conditions to avoid such *degenerative* solutions, which in turn aids in selecting allocator parameters.

To simplify the problem, let us consider the case $n_\tau = n_u = 1$. Most of the insights should (loosely) generalize to higher dimensions; details are left for future work. In this simplified case, $\mathbf{B} = b \in \mathbb{R}$, the uncertainty set is $\Delta_B = \{b_0 + \delta\sigma \mid |\delta| \leq 1\}$, and we assume $|b_0|, \sigma > 0$. Finally, without loss of generality, we set $u_0 = 0$ and assume $\underline{u} \leq 0 \leq \bar{u}$.

Let us define

$$u^+ := \max\{u, 0\}, \quad u^- := \max\{-u, 0\}. \quad (15)$$

Clearly $u^+, u^- \geq 0$ and it is easy to verify that $u = u^+ - u^-$ and $|u| = u^+ + u^-$. Then the simplified problem can be written as a standard LP:

$$\min_{u^+, u^-, \lambda, s} \quad c_1 u^+ + c_1 u^- - c_2 \lambda + s \quad (16a)$$

$$\text{s.t.} \quad \mathbf{0} \leq \begin{bmatrix} u^+ & u^- & \lambda & s \end{bmatrix}^T \quad (16b)$$

$$\begin{bmatrix} 0 \\ 0 \\ -\bar{u} \\ \underline{u} \\ 0 \\ -1 \\ 0 \\ 0 \end{bmatrix} \leq \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ -b_0 - \sigma & b_0 - \sigma & \tau_c & 1 \\ b_0 - \sigma & -b_0 - \sigma & -\tau_c & 1 \end{bmatrix} \begin{bmatrix} u^+ \\ u^- \\ \lambda \\ s \end{bmatrix} \quad (16c)$$

We are interested in finding sufficient conditions so that the solution $u \neq 0$ when $\tau_c \neq 0$. This is equivalent to finding necessary conditions for $u = 0$, and will lead to bounds on c_1 , c_2 , and σ , relative to b_0 and $|\tau_c|$.

We will need the following lemma, which is a special case of a standard LP result, but we prove it here for completeness.

Lemma 1 *Consider the simplified 1-dimensional RDPCP problem, Eqs. (16a)–(16c). If the optimal solution has $u^+ = u^-$, then $u^+ = u^- = 0$.*

Proof: Suppose for the sake of contradiction that $u^+ = u^- > 0$ is optimal. Then the last two inequality constraints in Eq. (16c) are

$$\begin{aligned} 0 &\leq -2\sigma u^+ + \lambda \tau_c + s \\ 0 &\leq -2\sigma u^+ - \lambda \tau_c + s \end{aligned}$$

which are clearly also satisfied by $u^+ = u^- = 0$. Furthermore, the cost function evaluated at $u^+ = u^- > 0$ is

$$\begin{aligned} J(u^+, u^+, \lambda, s) &= 2c_1 u^+ - c_2 \lambda + s \\ &> -c_2 \lambda + s \\ &= J(0, 0, \lambda, s). \end{aligned}$$

which implies that $u^+ = u^- > 0$ is suboptimal. $\square$

We can now derive necessary conditions under which $u = 0$ in the 1-dimensional setting.

Theorem 1 (Degenerative solutions of the 1-dimensional RDPCP problem) *Consider the simplified 1-dimensional RDPCP problem, Eqs. (16a)–(16c). The optimal solution has $u = u^+ - u^- = 0$ only if*

$$\min\{|\tau_c|, c_2\} \leq |\tau_c| \frac{c_1 + \sigma}{|b_0|}. \quad (17)$$

Proof: Suppose that the solution to Eqs. (16a)–(16c) has $u = u^+ - u^- = 0$. Then by Lemma 1, $u^+ = u^- = 0$. To derive necessary conditions for optimality, consider the dual problem (see [17, Theorem 1.3.2]) to Eqs. (16a)–(16c):

$$\max_{\mu \in \mathbb{R}^8} \quad -\bar{u} \cdot \mu_3 + \underline{u} \cdot \mu_4 - \mu_6 \quad (18a)$$

$$\text{s.t.} \quad \mathbf{0} \leq \mu \quad (18b)$$

$$\begin{bmatrix} c_1 \\ c_1 \\ -c_2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 & 0 & 0 & 0 & -b_0 - \sigma & b_0 - \sigma \\ 0 & 1 & 0 & -1 & 0 & 0 & b_0 - \sigma & -b_0 - \sigma \\ 0 & 0 & 0 & 0 & 1 & -1 & \tau_c & -\tau_c \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \mu \quad (18c)$$

A pair of feasible solutions to the primal and dual problems is optimal if and only if they satisfy the complementary slackness conditions [17, Theorem 1.3.3],

$$0 = \mu_1 u^+ \quad (19a)$$

$$0 = \mu_2 u^- \quad (19b)$$

$$0 = \mu_3 (\bar{u} - u^+) \quad (19c)$$

$$0 = \mu_4 (-\underline{u} - u^-) \quad (19d)$$

$$0 = \mu_5 \lambda \quad (19e)$$

$$0 = \mu_6 (1 - \lambda) \quad (19f)$$

$$0 = \mu_7 [(-b_0 - \sigma)u^+ + (b_0 - \sigma)u^- + s + \lambda\tau_c] \quad (19g)$$

$$0 = \mu_8 [(b_0 - \sigma)u^+ + (-b_0 - \sigma)u^- + s - \lambda\tau_c] \quad (19h)$$

and the zero duality gap condition [17, Theorem 1.3.3],

$$c_1 u^+ + c_1 u^- - c_2 \lambda + s = -\bar{u} \cdot \mu_3 + \underline{u} \cdot \mu_4 - \mu_6. \quad (20)$$

Since by Lemma 1, $u^+ = u^- = 0$, Eqs. (19a)–(19h) reduce to

$$0 = \mu_3 \bar{u} \quad (21a)$$

$$0 = \mu_4 \underline{u} \quad (21b)$$

$$0 = \mu_5 \lambda \quad (21c)$$

$$0 = \mu_6 (1 - \lambda) \quad (21d)$$

$$0 = \mu_7 (s + \lambda\tau_c) \quad (21e)$$

$$0 = \mu_8 (s - \lambda\tau_c) \quad (21f)$$

with μ_1, μ_2 free. Furthermore, using $\mu_3\bar{u} = \mu_4\underline{u} = 0$, Eq. (20) becomes

$$-c_2\lambda + s = -\mu_6. \quad (22)$$

Now we simplify Eq. (18c) to get

$$\mu_1 = c_1 + \sigma + b_0(2\mu_7 - 1) \quad (23a)$$

$$\mu_2 = c_1 + \sigma + b_0(-2\mu_7 + 1) \quad (23b)$$

$$\mu_5 - \mu_6 = \tau_c - c_2 - 2\tau_c\mu_7 \quad (23c)$$

$$\mu_8 = 1 - \mu_7 \quad (23d)$$

Using $\mu_1, \mu_2 \geq 0$, Eqs. (23a) and (23b) give us

$$-b_0(2\mu_7 - 1) \leq c_1 + \sigma, \quad b_0(2\mu_7 - 1) \leq c_1 + \sigma. \quad (24)$$

Together, these inequalities can be written as

$$|2\mu_7 - 1| \leq \frac{c_1 + \sigma}{|b_0|}. \quad (25)$$

Next, from Eq. (23d) we know that μ_7 and μ_8 cannot both be zero, therefore Eqs. (21e) and (21f) require that at least one of $s + \lambda\tau_c = 0$ or $s - \lambda\tau_c = 0$ holds. This is equivalent to requiring that at least one of $s + \lambda|\tau_c| = 0$ or $s - \lambda|\tau_c| = 0$ holds. Then since $s, \lambda \geq 0$, in either case it follows that

$$s = \lambda|\tau_c|. \quad (26)$$

Substituting this into Eq. (22) gives

$$\mu_6 = \lambda(c_2 - |\tau_c|). \quad (27)$$

Notice that because $\mu_6 \geq 0$, Eq. (27) implies that $\lambda = 0$ and/or $c_2 \geq |\tau_c|$. We consider these two cases separately.

In the first case, $\lambda = 0$, Eq. (27) immediately gives $\mu_6 = 0$. Substituting this into Eq. (23c) and recalling that $\mu_5 \geq 0$ yields

$$0 \leq \mu_5 = \tau_c - c_2 - 2\tau_c\mu_7 \implies c_2 \leq \tau_c(1 - 2\mu_7). \quad (28)$$

Since $c_2 > 0$, this is equivalent to $c_2 \leq |\tau_c|(1 - 2\mu_7)|$. Then applying Eq. (25) we obtain

$$\min\{|\tau_c|, c_2\} \leq c_2 \leq |\tau_c|(1 - 2\mu_7)| \leq |\tau_c| \frac{c_1 + \sigma}{|b_0|}, \quad (29)$$

which is the desired necessary condition for the case $\lambda = 0$.

Now suppose $c_2 \geq |\tau_c|$. If $\lambda = 0$, then we are done, so we can assume that $\lambda > 0$. Then we see that Eqs. (21e) and (21f) together with Eq. (26) become

$$0 = \mu_7(|\tau_c| + \tau_c) \quad (30a)$$

$$0 = \mu_8(|\tau_c| - \tau_c) \quad (30b)$$

If $\tau_c = 0$, then Eq. (17) holds trivially. We claim that for $|\tau_c| > 0$, we have $|2\mu_7 - 1| = 1$. To see this, notice that if $\tau_c > 0$, then $0 = |\tau_c|(2\mu_7)$, so $\mu_7 = 0$ and $|2\mu_7 - 1| = 1$. Similarly if $\tau_c < 0$, we have $0 = |\tau_c|(2\mu_8)$ so $\mu_8 = 0$. Then by Eq. (23d), $|2\mu_7 - 1| = |1 - 2\mu_8| = 1$. Hence Eq. (25) implies

$$\min\{|\tau_c|, c_2\} \leq |\tau_c| \leq |\tau_c| \frac{c_1 + \sigma}{|b_0|}, \quad (31)$$

as desired. Therefore the solution $\begin{bmatrix} u^+ & u^- & \lambda & s \end{bmatrix} = \begin{bmatrix} 0 & 0 & \lambda & s \end{bmatrix}$ is optimal only if Eq. (17) holds. $\square$

Theorem 1 immediately leads to the following constructive result, which provides conditions under which we can avoid degenerative solutions to Eqs. (16a)–(16c).

Corollary 1 Consider the simplified 1-dimensional RDPCP problem, Eqs. (16a)–(16c), with $|\tau_c| > 0$. If

$$1 > \frac{c_1 + \sigma}{|b_0|} \quad \text{and} \quad c_2 > |\tau_c| \frac{c_1 + \sigma}{|b_0|}, \quad (32)$$

then the optimal solution satisfies $u = u^+ - u^- \neq 0$.

Corollary 1 has a simple interpretation. It provides two conditions that, if satisfied, ensure the solution will not decay. First, to make the problem feasible, the nominal control effectiveness must be greater than the maximum uncertainty (plus c_1). If this basic condition is not satisfied, then the realized moment could possibly be in the opposite direction (sign) of the commanded moment. Second, it suffices to choose c_2 proportional to $|\tau_c|$, with the smallest permissible ratio of $c_2/|\tau_c|$ dictated by the amount of uncertainty. That is, take $c_2 = k|\tau_c|$ for a constant $k \in \mathbb{R}$ satisfying $k > (c_1 + \sigma)/|b_0|$. The same behavior has been qualitatively observed in higher-dimensional problems, but a proof deriving specific bounds is left for future work.

IV. Numerical Example

In this section we numerically study the efficacy of PI, MV, ℓ_1 LP, $R\ell_1$ CP, DPLP, and RDPCP allocators, and their robustness to model error. For this example we consider the linearized lateral-directional dynamics of the NASA F-18 High-Alpha Research Vehicle (HARV) [18, 19], shown in Fig. 1.



Fig. 1 NASA F-18 HARV in flight (credit: NASA Armstrong Flight Research Center).

In this example we command two generalized moments (angular accelerations),

$$\tau = \begin{bmatrix} \dot{p} & \dot{r} \end{bmatrix}^T, \quad (33)$$

where p and r are the body-axis roll and yaw rates. The control vector is

$$u = \begin{bmatrix} u_{DT} & u_A & u_R & u_{RTV} & u_{YTV} \end{bmatrix}^T, \quad (34)$$

with position limits listed in Table 1. Rate limits are ignored in this open-loop example, but can easily be incorporated if desired [1].

The nominal control effectiveness matrix is evaluated at Mach 0.2, an altitude of 10,000 ft, and angle of attack $\alpha = 29.73^\circ$ [19]:

$$B_0 = \begin{bmatrix} 1.879 & 1.328 & 0.02922 & 0.6754 & 0.217 \\ -0.1092 & -0.09645 & -0.08404 & 0.006811 & -2.974 \end{bmatrix} \quad (35)$$

Notice that the rudder is relatively ineffective at this high angle of attack.

Table 1 F-18 HARV aircraft control position limits based on Refs. [18, 19]

Control Input	Position Limit (deg)
Differential Tail (u_{DT})	± 17
Ailerons (u_A)	± 25
Rudder (u_R)	± 30
Roll Thrust Vectoring (u_{RTV})	± 25
Yaw Thrust Vectoring (u_{YTV})	± 25

We assume each individual entry of $\mathbf{B}$ is subject to independent Gaussian uncertainty, with standard deviations as shown in Table 2. The total uncertainty is the sum of these independent uncertainties. Standard deviations for aerodynamic surfaces, u_{DT} , u_A , and u_R , are based on uncertainties from [19, Table 5.1]. As Ref. [19] does not provide uncertainty for thrust vector controls, u_{RTV} and u_{YTV} , these are notional with scales chosen to be similar to the aerodynamic surface uncertainties.

Table 2 Standard deviations of control derivatives for the F-18 HARV example

	u_{DT}	u_A	u_R	u_{RTV}	u_{YTV}
$\dot{p}$	7.5%	5%	20%	5%	20%
$\dot{r}$	7.5%	10%	7.5%	20%	5%

We use unweighted versions of PI and MV allocators. A proper choice of weight matrix can be important to a practical design; in this case we use unweighted versions to illustrate the different qualitative behaviors of each allocator. Note that ℓ_1 LP, $R\ell_1$ CP, and RDPCP can also be tuned to minimize *weighted* ℓ_1 errors to prioritize certain moment axes when commands are unattainable.

Parameters for the ℓ_1 LP, $R\ell_1$ CP, and RDPCP allocators used for this example are reported in Eqs. (36) and (37). The same choice of c_1 (the weight on control effort minimization) is used for all allocators. The choice of c_2 for RDPCP is based heuristically on Corollary 1; a lower bound for c_2 in the multi-dimensional case will be derived in future work. For $R\ell_1$ CP and RDPCP, we use the independent ellipsoid uncertainty model in Eq. (6) with uncertainty covering two standard deviations from Table 2 (e.g., $\sigma_{1,3} = 2 \times 20\%[\mathbf{B}_0]_{1,3} = 0.0117$).

$$c_1 = 10^{-5} \|\mathbf{B}_0\|_1 = 3.2 \times 10^{-5} \quad (36)$$

$$c_2 = \|\boldsymbol{\tau}_c\|_1 \sum_{i=1}^{n_\tau} \frac{c_1 + \sqrt{\sum_{j=1}^{n_u} \sigma_{i,j}^2}}{\|[\mathbf{B}_0]_i\|_2} = 0.2386 \|\boldsymbol{\tau}_c\|_1 \quad (37)$$

A. Attainable Moment Sets and Nominal Allocation Accuracy

To evaluate allocator performance, begin by defining the Theoretical Attainable Moment Set (TAMS) as the set of all possible achievable moments due to all possible control effector inputs with nominal control effectiveness ($\mathbf{B} = \mathbf{B}_0$). Note that although in this section we evaluate results with $\mathbf{B} = \mathbf{B}_0$, the robust allocators still incorporate uncertainty in their allocation decision. Define the Particular Attainable Moment Set (PAMS) as the subset of the TAMS which can be realized by the specific allocator. The *allocation efficiency* is defined as the ratio of the volume of the PAMS to the TAMS [20]. If the allocator can realize the entire space in the TAMS then the allocator is perfectly efficient.

The efficiency of each allocator is reported in Table 3. Consistent with observations in Refs. [8, 21], the MV allocator is the least efficient as it is very conservative. Also consistent with prior research, the PI allocator has limited efficiency, whereas the optimization-based allocators can realize a much greater percentage of the TAMS. For this example problem, the robust allocators have practically the same efficiency as their standard counterparts.

Next, we characterize the magnitude and direction errors across a range of moment commands which contains the TAMS, $\boldsymbol{\tau}_c \in [-120, 120] \times [-100, 100] \text{ deg/s}^2$. Magnitude and direction errors for the nominal case ($\mathbf{B} = \mathbf{B}_0$) are

Table 3 Allocation efficiency (volume of PAMS / volume of TAMS) for the F-18 HARV example

Allocator	PI	MV	ℓ_1 LP	$R\ell_1$ CP	DPLP	RDPCP
Efficiency (%)	59	7	98	98	98	97

shown in Figs. 2 and 3, respectively. Total control utilization as a percentage of available control is shown in Fig. 4. These plots also include the TAMS and the approximate PAMS of each allocator, which allows one to visualize each allocator's efficiency. In the nominal case, the behavior *outside* the PAMS is of particular interest, as this can have significant impacts on stability, performance, and handling qualities.

First, observe the PI and MV allocators have zero direction error both inside and outside the TAMS. This is because the effectors are saturated proportionally to the most saturated effector, allowing moment direction to be preserved. The PI and MV allocators also exhibit the desirable property of scaling linearly in magnitude error outside its PAMS. The same holds for the DPLP and RDPCP allocators, but DPLP and RDPCP are much more efficient than PI and MV.

As expected, ℓ_1 LP and $R\ell_1$ CP have the lowest magnitude error for all allocators, as they are specifically formulated to minimize the ℓ_1 error. Outside the TAMS, direction errors of ℓ_1 LP and $R\ell_1$ CP increase noticeably. In contrast, DPLP and RDPCP have somewhat greater magnitude error, but this error scales linearly with the command magnitude, and the tradeoff is necessary to minimize direction error.

Finally, in comparison to standard DPLP, the proposed RDPCP allocator has reduced control utilization both inside and outside the PAMS. This is due to the bias towards the preferred control (zero) and the robustified constraints. Reduced control usage also helps maintain stability margins and control authority, and reduces actuator wear. Similar behavior is observed for $R\ell_1$ CP compared to standard ℓ_1 LP.

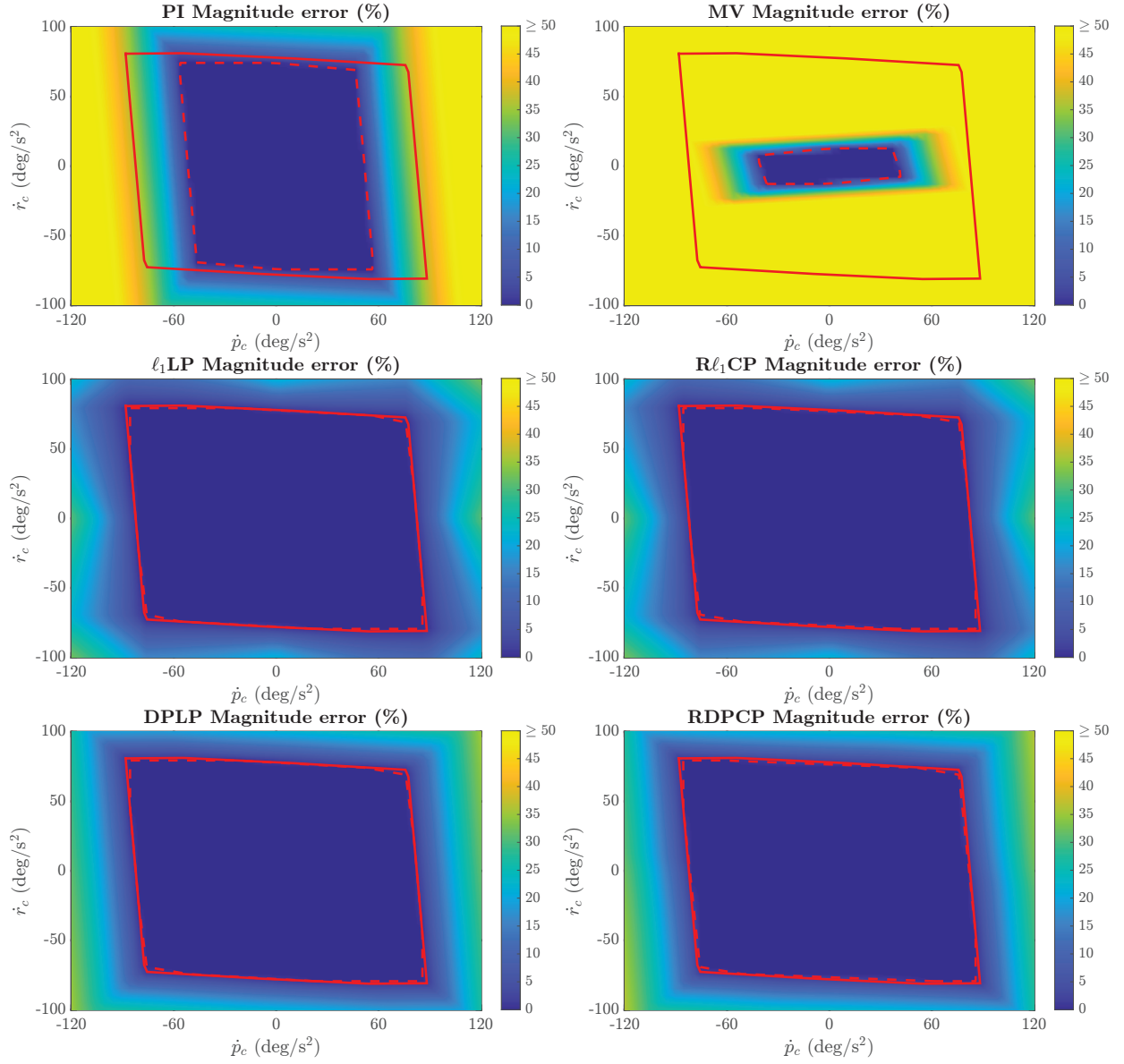


Fig. 2 Relative magnitude errors for each allocator under nominal conditions. The TAMS is shown as a solid red line, and the approximate PAMS as a dashed red line.

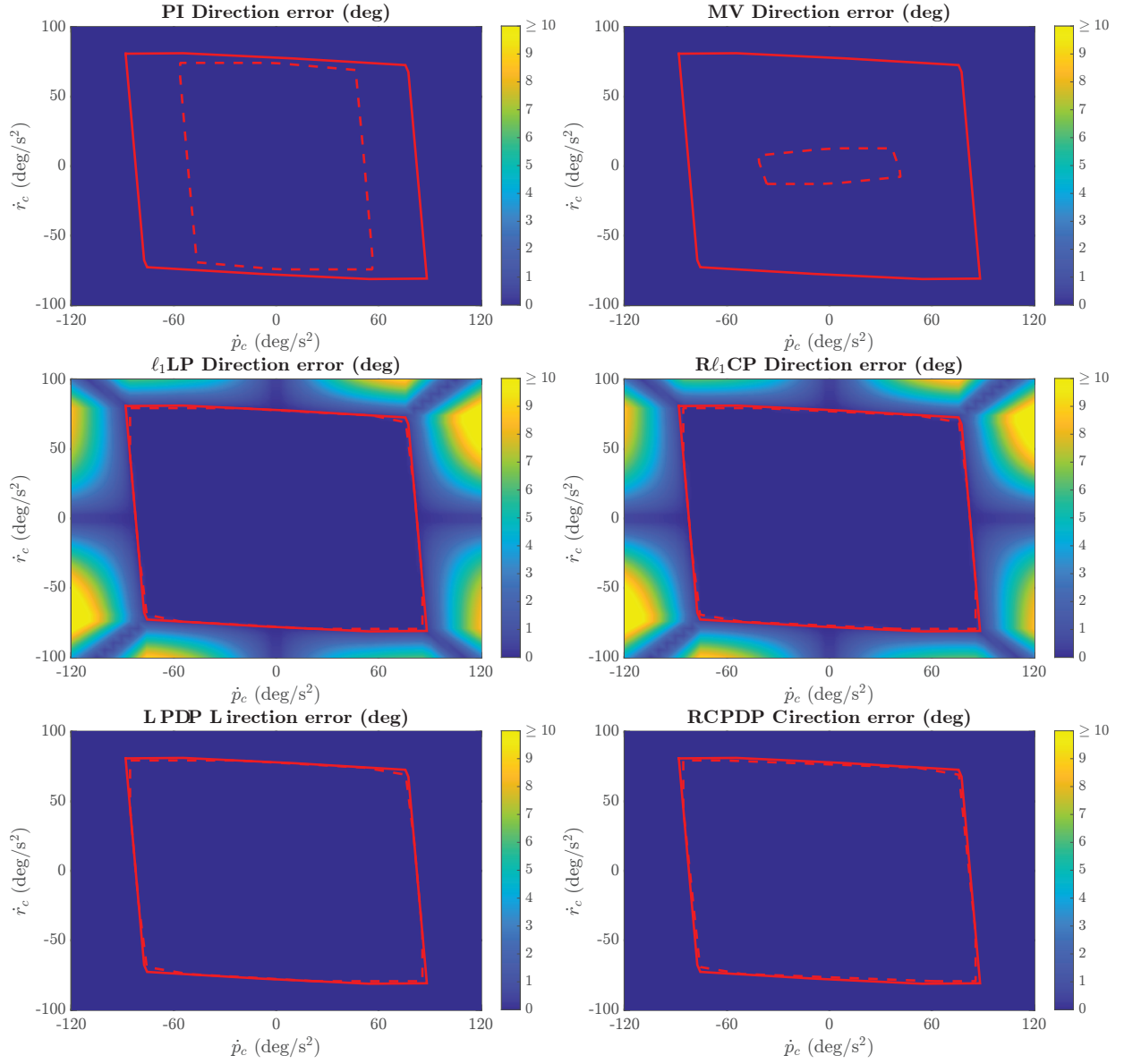


Fig. 3 Relative direction errors for each allocator under nominal conditions. The TAMs is shown as a solid red line, and the approximate PAMs as a dashed red line.

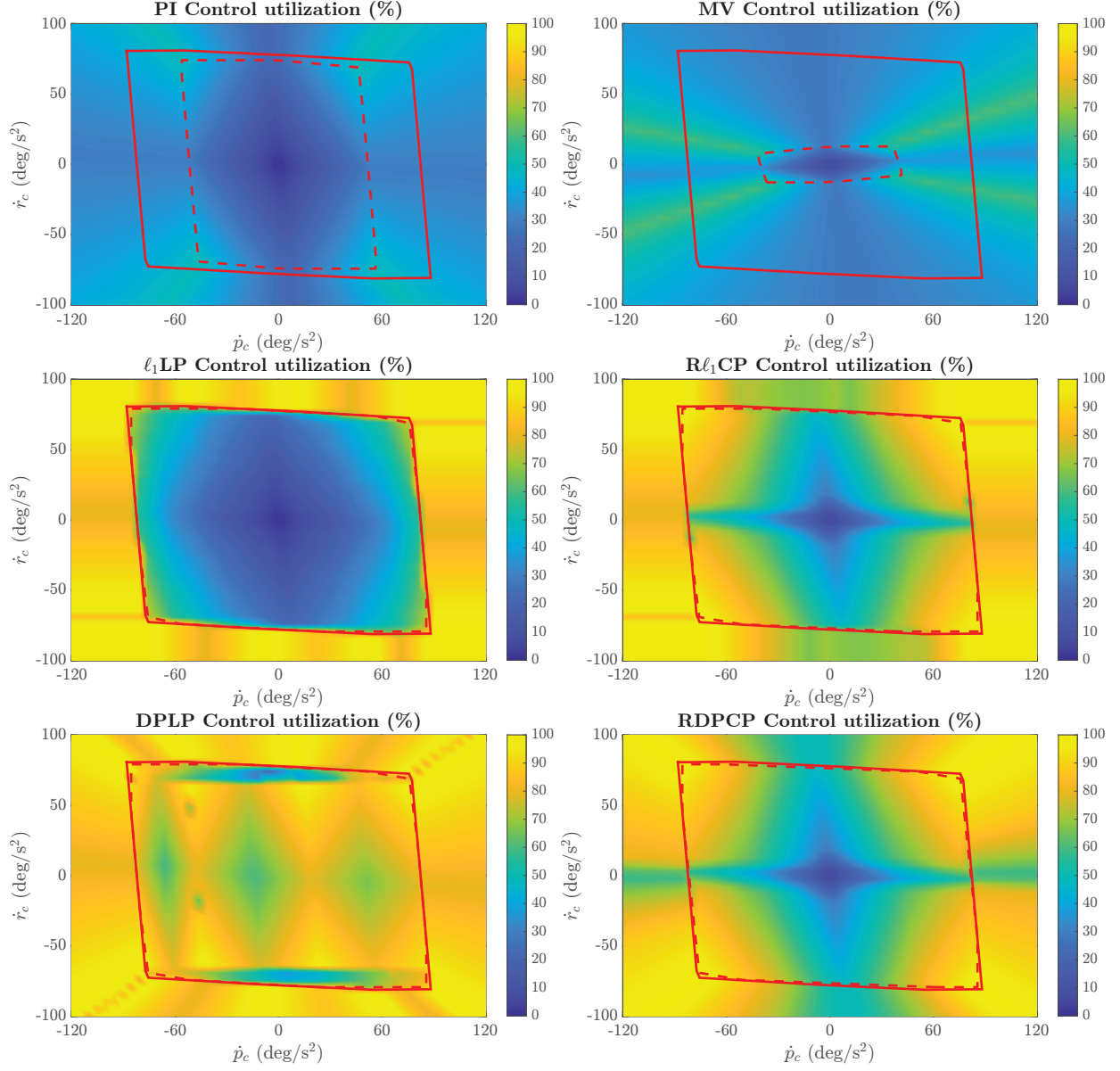


Fig. 4 Total control utilization, $2\|\mathbf{u}\|_1/\|\bar{\mathbf{u}} - \underline{\mathbf{u}}\|_1$, for each allocator. The TAMS is shown as a solid red line, and the approximate PAMS as a dashed red line.

B. Allocation Accuracy under Uncertainty

Now we evaluate the robustness of each allocator subject to independent Gaussian variations in entries of $\mathbf{B}$ as specified in Table 2. We evaluate this robustness for a set of representative commands just outside the TAMS. Distributions of realized moments under this dispersion in $\mathbf{B}$ are shown in Fig. 5. Cumulative distribution functions (CDFs) of the associated magnitude and direction of the allocation errors for three illustrative cases are shown in Fig. 6.

In Figs. 5 and 6 we immediately see that the MV allocator is exceedingly conservative. While it indeed achieves the smallest variance and robustly small direction error, this comes at the cost of an unacceptable level of magnitude attenuation. This attenuation was already observed under nominal conditions (see Fig. 2) and is the result of MV relying on control inputs with the smallest variance. This behavior can be affected by modifying the MV weight matrix [8].

Similarly, PI allocation solutions also have small variance and robustly small direction error. However, due to its inability to access the full TAMS (see also Fig. 2), PI suffers from relatively large magnitude error for these moment commands and is not robust in this respect.

Compared to PI and MV, the optimization-based allocators were all able to attain moments much closer to the command. As expected, ℓ_1 LP had smaller magnitude error than DPLP, whereas DPLP had smaller direction error (as indicated by their respective CDFs lying above and to the left).

Lastly, compared to the standard LP allocators, the robust SOCP allocators ($R\ell_1$ CP and RDPCP) had smaller variance and accomplished their respective optimization objectives more robustly. This enhanced robustness is especially notable when a large yaw rate command ($\dot{r}_c$) is paired with a small roll rate command ($\dot{p}_c$). Such commands are important for executing stability-axis roll maneuvers at high angles of attack. These results demonstrate that RDPCP can deliver more robust moment direction preservation than a standard DPLP allocator, especially at the edge of the envelope.

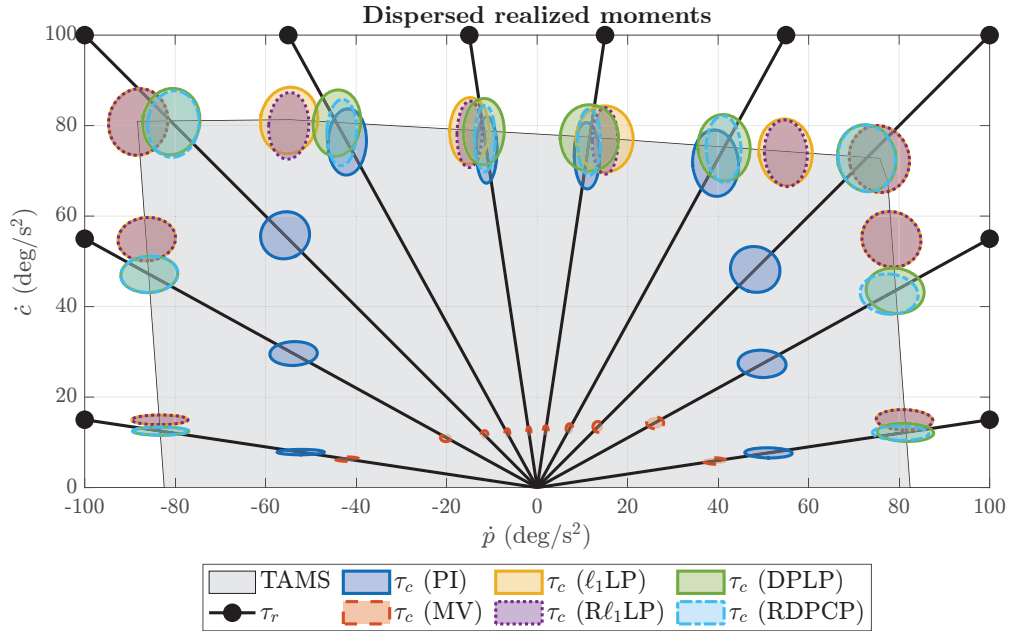


Fig. 5 2-sigma covariance ellipses for realized moments with Gaussian dispersions in $\mathbf{B}$.

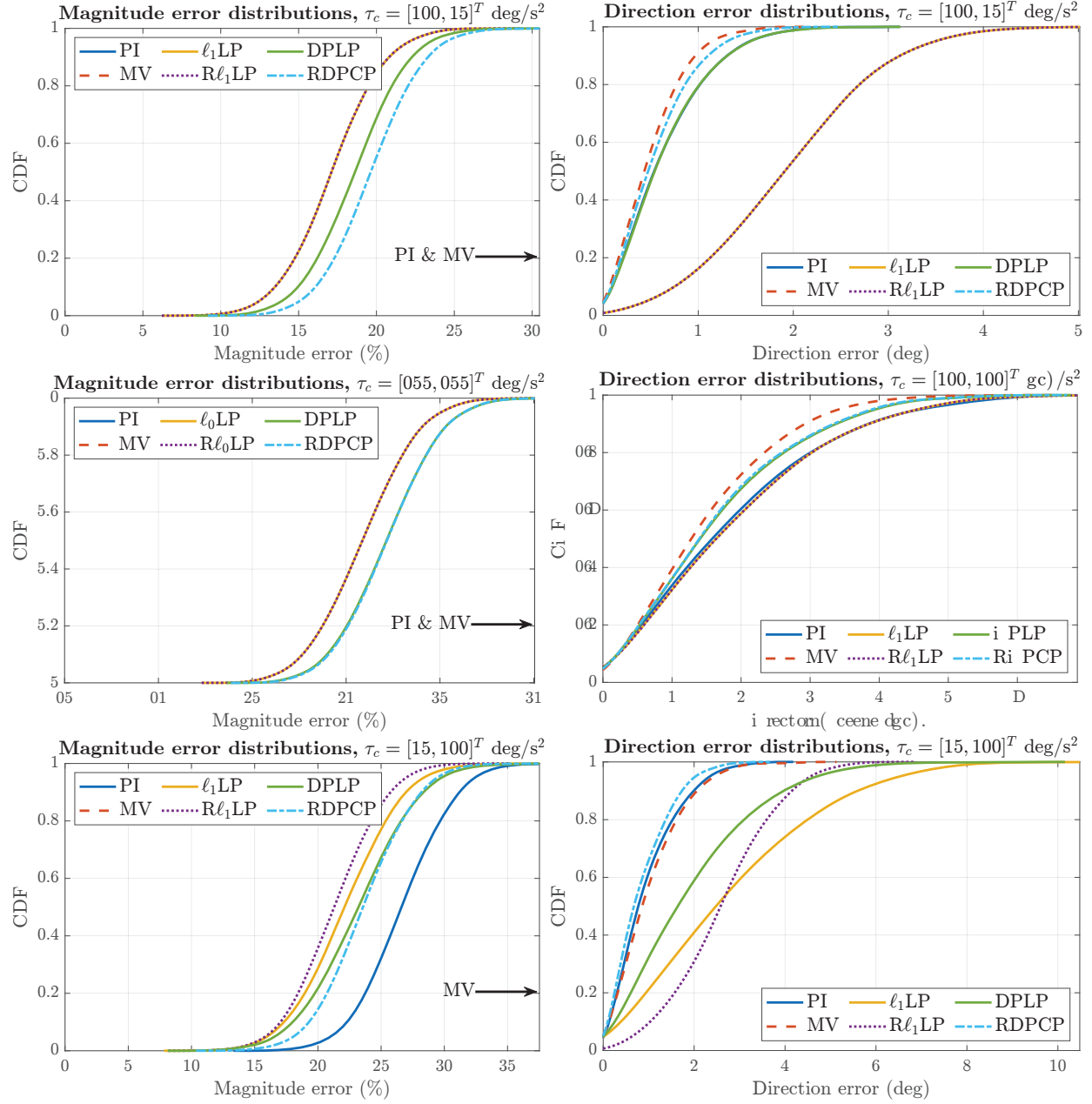


Fig. 6 CDFs of magnitude errors (left) and direction errors (right) of realized moments with Gaussian dispersions in B , for a selection of moment commands. On these plots, CDFs which are further left indicate more robustly accurate allocation solutions. Some magnitude error CDFs for PI and MV are excluded as they are much larger than the other allocators.

C. Computational Efficiency

For practical application, allocation schemes must be able to deliver solutions in real time. GI allocators (e.g., PI and MV) require only one matrix inversion, which is done offline, but optimization-based methods must be run online for each new moment command. To assess the feasibility of the proposed RDPCP allocator compared to ℓ_1 LP, DPLP, and $R\ell_1$ CP, Fig. 7 shows the distribution of computational costs for generating the allocation solutions to the F-18 HARV example problem reported in section IV.A. Solution times are for an implementation in MATLAB[®] R2024b run on a 2023 Apple[®] MacBook Pro with M2 Pro chip and 16 GB RAM. LPs are solved using MATLAB[®]'s `linprog` and SOCPs with `coneprog`. An optimized implementation in compiled code would, of course, be considerably faster [5].

Figure 7 confirms that the standard LP-based allocators are quite fast, with a median CPU time of about 0.3ms per solution. The robust SOCP-based allocators are somewhat slower, with a median CPU time of about 1ms per solution—but this is still fast enough for real-time implementation. We also highlight that the proposed RDPCP is similarly fast to $R\ell_1$ CP, so the the control design can prioritize preserving moment directions without sacrificing computational efficiency.

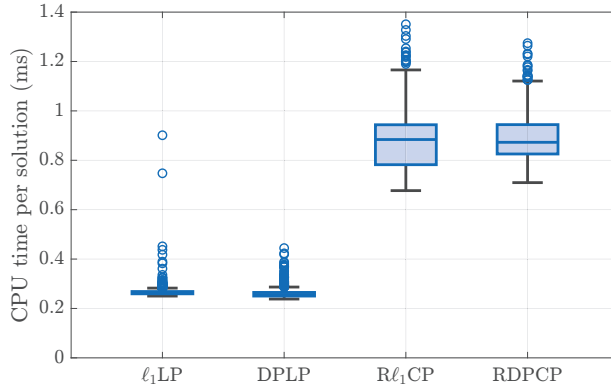


Fig. 7 Distribution of computational costs (ms/solve) for optimization-based allocators applied to the F-18 HARV example problem.

V. Conclusions

In this paper we have described a control allocation scheme which is designed to robustly preserve the direction of generalized moment commands under bounded, parametric model uncertainty. The proposed approach, RDPCP, can be viewed as a generalization of $R\ell_1$ CP, and like this other method, is based on solving a SOCP. Therefore, it can realize virtually all attainable moments and is suitable for real-time implementation. By tuning the parameters of the SOCP, RDPCP can emphasize robust moment direction preservation, robust minimum allocation error, or minimum control effort, while being able to realize the full TAMS. Depending on the control allocation problem, these properties can in turn allow for a robust and flexible design which may mitigate undesired cross-axis coupling and improve stability margins and handling qualities.

In addition, for the one-dimensional case, we have derived bounds on the allocator parameters which guarantee the solution will not be degenerative (decay to zero even for non-zero moment commands). Future research should aim to extend this analysis to the general multi-dimensional case. Finally, we have compared RDPCP to other allocation schemes applied to the lateral-directional dynamics of the F-18 HARV. In open-loop tests, RDPCP recovered the allocation efficiency of other LP-based allocation schemes while robustly preserving moment direction inside and outside the TAMS. Additional work is needed to conduct closed-loop and piloted simulations to evaluate any potential advantages or tradeoffs in stability, robustness, and handling qualities.

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